

Multilingual Hate Speech and Slang Detection: Evaluating Hybrid Transformers Against Zero-Shot and Few-Shot LLMs Struggling with Short and Low-Context Texts

Abstract—The increasing prevalence of multilingual social media platforms has intensified the challenges of detecting hate speech and abusive slang, particularly when addressing both high-resource languages such as English and low-resource languages such as Bangla. Although large language models (LLMs) demonstrate promising zero-shot and few-shot capabilities, they frequently encounter difficulties with short, ambiguous, and slang-rich content when compared to smaller, domain-specific transformer models. This study provides a comprehensive evaluation of toxic and suspicious language detection using two benchmark datasets: Toxic Tweets (English) and Suspicious Bangla Text (Bangla). A hybrid classification framework is introduced, integrating fine-tuned BERT architectures with zero-shot and few-shot LLM prompting strategies. Three pipelines are investigated—supervised BERT encoding, zero-shot classification with GPT-OSS 120B, and retrieval-augmented few-shot prompting—and evaluated using precision, recall, F1-score, and accuracy. Experimental results indicate that the specialized BERT-based classifier achieves the highest performance, with an accuracy of 90% for Bangla and 91.1% for English, surpassing both zero-shot and few-shot LLM approaches. These findings emphasize the persistent performance gap between general-purpose LLMs and task-specific models in multilingual hate speech detection, highlighting the importance of domain adaptation and hybrid modeling for effective moderation in linguistically diverse online environments.

Index Terms—Hate Speech Detection, Bangla Slang, BERT, LLM, Zero-Shot, Few-Shot, Toxicity Classification.

I. INTRODUCTION

Hate speech is broadly defined as communication that attacks or demeans individuals or groups based on attributes such as race, religion, or gender [1]. Online, it often appears in implicit or coded forms, complicating detection. In parallel, suspicious Bangla text refers to abusive slang, informal expressions, and transliterated phrases common on social media. These are difficult to identify due to the language’s rich morphology, frequent code-switching, and continual emergence of new slang [2].

Detection is challenging both linguistically and computationally. English hate speech and Bangla slang rely on subtle cues, sarcasm, abbreviations, and fast-evolving vocabulary. At the same time, annotated datasets remain scarce, especially for Bangla, limiting robust model training [3]. Unbalanced corpora further risk amplifying bias, leading to unfair moderation outcomes.

Researchers have mainly treated this as a classification task. Early machine learning methods, including logistic regression and Support Vector Machines (SVMs), provided baselines but failed to capture contextual nuance [4]. Transformer-based encoders, notably BERT, marked a major advance by capturing meaning beyond surface-level words [5]. Fine-tuned BERT models, sometimes combined with recurrent or convolutional layers, achieve strong performance with about 150M parameters, offering effectiveness and efficiency [6]. For Bangla, detection tasks are typically binary ($\{0, 1\}$: neutral vs. suspicious), while English datasets often use three classes ($\{0, 1, 2\}$: hate, offensive, neutral).

Compared to these specialized models, large language models (LLMs) such as GPT, LLaMA, and Mistral, with up to 70B parameters, show remarkable zero-shot and few-shot capabilities [7]. Yet, they often underperform domain-specific models in hate speech and slang detection [8], since LLMs are optimized for generation rather than classification. They misclassify short, ambiguous, slang-heavy utterances requiring fine-grained contextual interpretation [9].

Addressing this limitation requires inclusive, slang-aware datasets, transformer fine-tuning for domain adaptation, and fairness-aware objectives [10]. Effective detection fosters safer online environments, protects vulnerable communities, and ensures trustworthy moderation. Moreover, adapting LLMs through fine-tuning or hybrid pipelines may extend their utility, harnessing generative power while compensating for classification weaknesses.

This raises a central question: why do smaller models like BERT often outperform massive LLMs in Bangla slang detection, and how can this gap be bridged? To answer, this study compares open-source LLMs with BERT-based architectures on English and Bangla datasets, evaluating accuracy, F1-score, and fairness. The results identify effective models and demonstrate how domain-specific fine-tuning with scalable LLMs can advance multilingual toxic content moderation.

II. LITERATURE REVIEW

Research on hate speech detection in English and suspicious Bangla slang has progressed along parallel yet complementary directions. Early classifiers, including logistic regression, decision trees, and Support Vector Machines (SVMs), were

The advent of deep learning and transformer-based architectures marked a turning point. In English, models such as BERT and RoBERTa captured contextual nuance, outperforming earlier baselines on benchmark datasets [12]. In Bangla, annotated corpora such as BIDWESH [13], the Bangla Hate Speech datasets [14], [15], and the Suspicious Bangla Slang corpus [16] enabled fine-tuned BERT-style models to excel at abusive slang detection [17]. Hybrid models like BERT-BiLSTM further improved accuracy in imbalanced settings [18].

This gap reflects architectural differences. BERT’s bi-directional encoder, trained via masked language modeling, is optimized for classification and fine-tuning. LLMs, as autoregressive decoders trained primarily for generation, treat classification as an emergent rather than optimized ability [22], [23]. Consequently, LLMs often misclassify short or ambiguous utterances, especially in Bangla slang where fine-grained cues dominate [18].

III. MATERIALS AND METHODS

Two publicly available datasets were used to examine toxic and suspicious language patterns in English and Bengali. The first, *Toxic Tweets*, sourced from Kaggle and X posts (formerly Twitter), is annotated for toxicity across categories such as abusive, hate speech, and non-toxic content, providing a key resource for analyzing harmful English social media communication [26].

[illegible]

	text	label
0	!!!! RT @mayaslovely: As a woman you shouldn't...	2
1	!!!!!! RT @lmeew17: boy dats cold...lyga dw ba...	1
2	!!!!!! RT @UrKind0Brand Dawg!!!! RT @80sbaby...	1
3	!!!!!! RT @C_G_Anderson @viva_based she lo...	1
4	!!!!!! RT @ShenikaRoberts: The shd you...	1
24776	you's a mutha!!!! in lie #8220, @LifeAsKing, @2...	1
24779	you've gone and broke the wrong heart baby, @2...	1
24780	young buck wanna aell... dat ngugh like i ain...	1
24781	you got wild bitch tellin you lie	1
24782	--Ruffed I Nae Eileen Dahlia - Beautiful col...	1

Together, these datasets provide a dual perspective: *Toxic Tweets* reflects global English toxicity patterns, while the Bengali corpus highlights challenges in low-resource and regional language contexts. Their combined use enables a comprehensive evaluation of detection systems across both high- and low-resource languages.

Unlike conventional NLP workflows that apply extensive normalization, this study deliberately avoids aggressive cleaning to preserve the raw structure of user-generated content and reflect real-world conditions. Social media texts often include spelling variations, emojis, code-mixing, and slang that add complexity to toxicity detection. Removing them risks discarding critical contextual and semantic cues. Representative samples from both datasets are shown in Fig. 1.

The proposed methodology begins with the raw Bangla and English datasets, where preprocessing and prompting serve as the foundational steps. Three complementary approaches are integrated: supervised BERT encoding, zero-shot LLM prompting, and few-shot LLM classification with retrieval.

In the supervised pipeline, text is tokenized and encoded using BERT, after which dense and softmax layers are applied for classification. In the zero-shot setting, prompts are passed directly to GPT-OSS 120 B, with outputs refined through regex-based parsing. The few-shot pathway extends this approach by retrieving top- k examples through FAISS embeddings, which are incorporated into the prompt before classification.

The workflow illustrated in Fig. 2 demonstrates the complete process. Following dataset preprocessing, the methodology diverges into three pipelines—supervised, zero-shot,

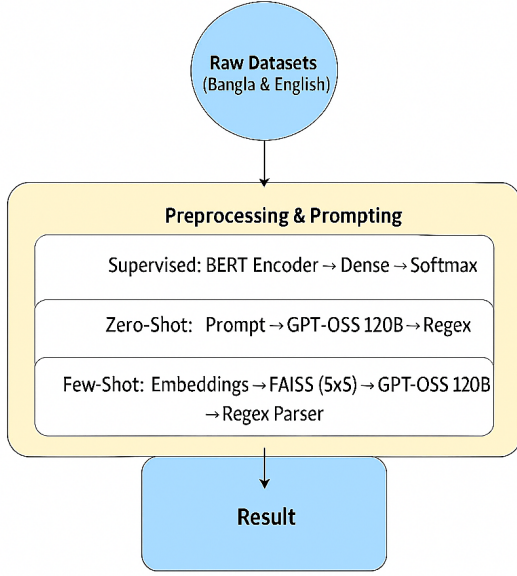


Fig. 2: Overview of the Proposed Methodology

and few-shot—each employing distinct modeling strategies. Predictions from these approaches are then consolidated and evaluated using standard metrics, including accuracy, precision, recall, and F1-score. By integrating multiple strategies, the hybrid framework ensures robust performance across languages and underscores its adaptability for multilingual suspicious and toxic text classification.

D. Model Architecture

1) *BERT-Based Classification Architecture*: The proposed model processes raw textual input, either English toxic tweets or suspicious Bengali slang, through a BERT backbone to extract contextual embeddings. Initially, the text is tokenized using HuggingFace’s AutoTokenizer, producing subword tokens compatible with the transformer encoder. The BERT encoder then generates a sequence of hidden states, with the mean of the final hidden layer employed as the sentence-level embedding:

$$h_{\text{avg}} = \frac{1}{n} \sum_{i=1}^n h_i \quad (1)$$

where h_i denotes the hidden state of the i -th token, and n is the total number of tokens.

This pooled representation is passed through a linear layer with ReLU activation, followed by dropout at a rate of 30% to introduce non-linearity and reduce overfitting:

$$z = \text{Dropout}(\max(0, W_1 h_{\text{avg}} + b_1)) \quad (2)$$

Finally, a softmax classification layer maps the resulting feature vector to output probabilities across the label space:

$$\hat{y} = \arg \max_{c \in \{1, \dots, C\}} \text{softmax}(W_2 z + b_2) \quad (3)$$

where C denotes the number of target classes, which is binary for suspicious versus non-suspicious detection in Bangla, and three-class for toxic, offensive, or neutral categorization in English.

2) *Zero-Shot LLM Classification*: A Large Language Model (LLM) is a transformer-based architecture trained on extensive corpora of textual data, enabling generalization across a wide range of tasks without explicit fine-tuning. In the zero-shot setting, the model relies solely on natural language instructions provided through a system prompt and directly predicts the label for an unseen input x . The prediction process can be expressed as:

$$\hat{y} = \arg \max_{c \in C} P(c | x, \text{prompt}), \quad (4)$$

where C represents the set of predefined labels and $\hat{y}$ denotes the predicted class. This formulation evaluates the model’s capacity to generalize effectively in the absence of task-specific training data, thereby serving as a benchmark for assessing zero-shot adaptability in multilingual toxicity detection.

3) *Few-Shot Classification with Retrieval-Augmented Generation*: The few-shot classification framework employs a retrieval-augmented strategy that allows a Large Language Model (LLM) to utilize contextual examples for improved classification performance. The architecture consists of four main stages: embedding generation, vector database construction, example retrieval, and LLM-based classification.

In the embedding generation stage, raw text inputs are processed through a BanglaBERT encoder to produce dense vector representations. The tokenization process converts each input into subword tokens with a maximum sequence length of 256, and the [CLS] token embedding is adopted as the sentence-level representation:

$$\mathbf{e} = \text{BanglaBERT}(\text{tokenize}(x))[1, :] \quad (5)$$

The encoded embeddings are then used to construct a vector database. Training examples are encoded and stored in FAISS (Facebook AI Similarity Search) to enable efficient similarity search. To support cosine similarity retrieval, embeddings are normalized using the L2 norm:

$$\mathbf{e}_{\text{norm}} = \frac{\mathbf{e}}{\|\mathbf{e}\|_2} \quad (6)$$

An IndexFlatIP index is created for exact similarity search across the training corpus.

For each test instance, the system retrieves the top- k most similar training examples based on cosine similarity:

$$\text{examples} = \arg \max_k \text{cosine_sim}(\mathbf{e}_{\text{query}}, \mathbf{e}_{\text{train}}) \quad (7)$$

In this study, $k = 5$ examples are retrieved to provide diverse and contextually relevant information.



Fig. 3: Proposed BERT-Based Classification Architecture

The retrieved examples are then incorporated into a structured prompt template and passed to the GPT-OSS 120B model. The system prompt includes the task definition and classification guidelines, descriptions of the labels (0: Hate Speech, 1: Offensive Language, 2: Neutral Content), the retrieved examples with their associated labels, and output format constraints requiring a single-digit response. The final classification is obtained through prompt-based inference:

$$\hat{y} = \text{LLM}(\text{prompt}(\text{examples}, x_{\text{test}})) \quad (8)$$

By combining the semantic richness of transformer embeddings with the reasoning capabilities of large language models, this retrieval-augmented few-shot framework enables robust multilingual classification, even in settings with limited task-specific supervision.

IV. EXPERIMENTAL ANALYSIS

A. Experimental Setup

1) *Computational Environment and Models:* All experiments were conducted using the Kaggle environment. Training and fine-tuning of the hybrid model were performed on a Tesla T4 GPU equipped with 16GB of VRAM, while inference tasks for the LLMs were executed on a standard Kaggle CPU instance. The study presents a comparative analysis between the proposed specialized classifier and general-purpose large language models. For the hybrid classifier, language-specific BERT variants were employed: csebuetnlp/banglabert for the Bengali dataset and bert-base-uncased for the English dataset. To provide a benchmark against state-of-the-art generalist models, experiments were conducted with the GPT-OSS 120B model, which offers representative performance for zero-shot and few-shot evaluation. A summary of the primary components of the computational environment and selected models is presented in Table I.

2) *Experimental Procedure:* The evaluation was organized into three phases to ensure a systematic comparison of methodologies. In the first phase, both Bengali and English datasets were classified using the fine-tuned hybrid BERT-based model. In the second phase, the same test sets were evaluated using the LLM in a zero-shot configuration, relying solely on natural language prompts without task-specific examples. In the third phase, the test sets were reclassified using the LLM with a few illustrative examples incorporated into the prompt, thereby leveraging in-context learning for improved performance. This structured design enables a direct comparison between supervised fine-tuning, zero-shot prompting,

and retrieval-augmented few-shot prompting, highlighting the relative strengths and limitations of each approach.

TABLE I: Specifications of pre-trained models, large language models, and hardware components employed in the experimental setup.

Component	Specification
Pre-trained Models	csebuetnlp/banglabert (110M) bert-base-uncased (110M)
Large Language Model	GPT-OSS (120B)
GPU Hardware	Tesla T4 (16GB)

B. Result and Evaluation

This section presents the performance outcomes obtained from each phase of the experiment. A consolidated summary of results is provided in Table II, which reports the evaluation metrics for all models across both datasets. The table demonstrates a clear performance hierarchy: the zero-shot configuration consistently produces the lowest scores, the few-shot configuration yields noticeable improvements, and the hybrid classifier achieves the highest accuracy with balanced outcomes across all metrics.

1) *Performance of the Hybrid Classifier:* The hybrid model was fine-tuned for five epochs on each training set and delivered consistently strong results. For the Bengali dataset, the model attained balanced scores, achieving Precision, Recall, and F1-score of 90%, along with an overall accuracy of 90%. For the English dataset, the model demonstrated similarly robust performance, with a Precision of 90%, a Recall of 91%, an F1-score of 90%, and an overall accuracy of 91.1%. These results are particularly noteworthy given the inherent challenges of the datasets, which include pronounced class imbalance and substantial textual overlap between categories.

2) *LLM Performance in a Zero-Shot Setting:* In the zero-shot evaluation, the GPT-OSS 120B model attained a Precision of 88%, a Recall of 76%, an F1-score of 80%, and an overall accuracy of 75.52% for the English dataset. For the Bengali dataset, the model achieved a Precision of 75%, a Recall of 74%, an F1-score of 74%, and an accuracy of 74.19%.

3) *LLM Performance with Few-Shot Learning:* By incorporating in-context learning, the LLM’s performance demonstrated marked improvement. For the English dataset, the few-shot model achieved a Precision of 89%, a Recall of 82%, an F1-score of 84%, and an accuracy of 81.16%. For the Bengali dataset, the model produced balanced outcomes, with Precision, Recall, and F1-score each at 82%, and an overall accuracy of 86.61%.

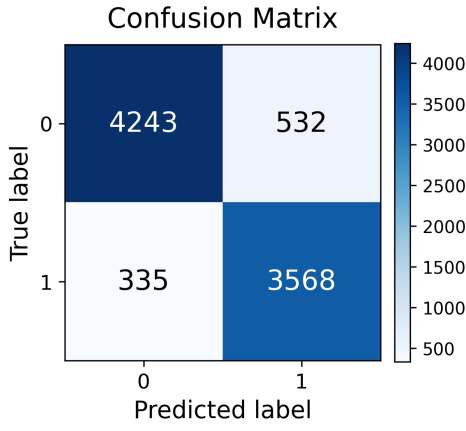


Fig. 4: Confusion matrix illustrating the classification outcomes of the hybrid model on the Bengali dataset.

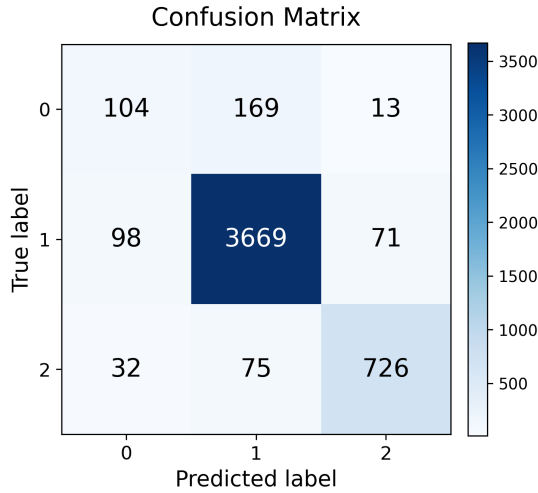


Fig. 5: Confusion matrix illustrating the classification outcomes of the hybrid model on the English dataset.

4) *Consolidated Results*: Table II and Fig. 4–5 together provide a comprehensive comparison of performance across all methods and datasets. The results clearly demonstrate a performance hierarchy, beginning with the baseline zero-shot evaluation, followed by substantial gains in the few-shot configuration, and culminating in the superior accuracy achieved by the specialized hybrid classifier.

V. RESULT ANALYSIS AND DISCUSSION

A. Superior Performance of the Specialized Hybrid Model

The results clearly demonstrate the efficacy of the proposed hybrid model, which integrates BERT with a Long Short-Term Memory (LSTM) network. Across both Bengali and English datasets, the hybrid classifier achieved the highest performance, with F1-scores of 90%. This superiority stems from task-specific specialization: fine-tuning on target datasets enables the model to capture nuances, vocabulary, and contextual patterns typical of noisy, short-text domains. Through fine-

TABLE II: Comparative performance of the hybrid classifier, zero-shot, and few-shot methods on Bengali and English datasets, reported in terms of Precision, Recall, F1-score, and Accuracy (%).

Dataset	Method	Precision	Recall	F1-Score	Accuracy
Bengali	Hybrid	90.00	90.00	90.00	90.00
	Zero-shot	75.00	74.00	74.00	74.19
	Few-shot	82.00	82.00	82.00	82.16
English	Hybrid	90.00	91.00	90.00	91.10
	Zero-shot	88.00	76.00	80.00	75.52
	Few-shot	89.00	82.00	84.00	81.61

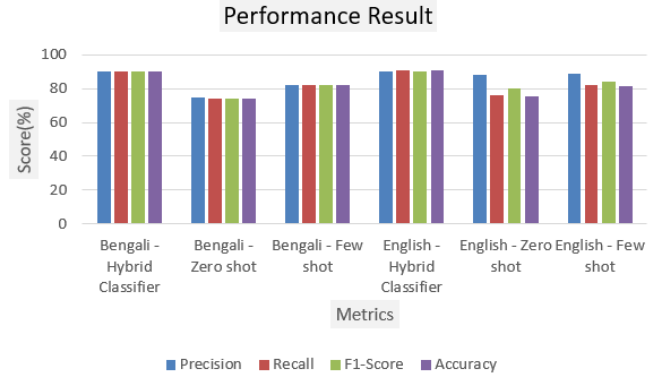


Fig. 6: Comparative visualization of Precision, Recall, F1-score, and Accuracy across hybrid, zero-shot, and few-shot methods for both Bengali and English datasets.

tuning, the hybrid architecture develops specialized linguistic representations, making it highly effective for this classification task.

B. Analyzing the Performance of Large Language Models (LLMs)

A central objective of this research was to benchmark general-purpose LLMs against the specialized hybrid model. The findings reveal a marked disparity, particularly in zero-shot settings.

Without task-specific examples, the LLM attained lower accuracy than the hybrid classifier, with 74.19% on Bengali and 75.52% on English. This gap does not signify failure but reflects its design philosophy. Trained on vast heterogeneous corpora, LLMs function as broad generalists, but this generality limits performance in specialized domains or data lacking rich contextual cues. As shown in prior work, performance is highly sensitive to alignment between task and pre-training distribution [27]. The datasets here, brief, informal, and weakly structured, exemplify domains where general-purpose models encounter significant constraints.

C. The Impact of In-Context Learning (Few-Shot Prompting)

Performance improved substantially when the LLM was shifted from zero-shot to few-shot prompting. Providing only a handful of examples increased accuracy on Bengali from 74%

to 82%. This illustrates the adaptability of in-context learning, which conditions responses more effectively and aligns outputs with task logic.

Nevertheless, the few-shot LLM did not surpass the hybrid model. While in-context learning is powerful for rapid adaptation, it cannot replace the domain-specific knowledge gained through dedicated fine-tuning. The limitation is most evident when classifying ambiguous or neutral content, where subtle contextual cues define category boundaries.

D. The Generalist–Specialist Trade-off and Future Directions

The analysis underscores a fundamental trade-off in NLP: generalist LLMs offer versatility across tasks without retraining, while specialist models excel in accuracy within complex, domain-specific contexts. For noisy, short-text classification, a fine-tuned specialist remains more reliable.

The progression from GPT-2 to contemporary multimodal systems shows steady advancement in LLM reasoning and task generalization. This study contributes to that trajectory by highlighting robustness in low-context scenarios as a persistent challenge. For universal applicability, future LLMs must integrate more sophisticated mechanisms for handling ambiguity and making accurate inferences from sparse or weakly structured input.

VI. CONCLUSION

This study provided a comprehensive evaluation of hybrid classification approaches for multilingual text processing, focusing on Bengali and English datasets. By integrating pre-trained transformer models with large language models, the proposed hybrid architecture delivered strong results, achieving accuracies of 90% for Bengali and 91.1% for English. The findings emphasize the advantages of specialization, as the hybrid model effectively captured domain-specific nuances and surpassed both zero-shot and few-shot LLM configurations. Zero-shot evaluation demonstrated the adaptability of general-purpose models for cross-lingual tasks but exposed weaknesses in low-resource languages. Few-shot prompting improved performance through in-context learning yet still fell short of the fine-tuned hybrid model. Overall, the results highlight the importance of combining generalist and specialist strategies to ensure robust multilingual classification in noisy, short-text contexts. This work contributes to multilingual NLP by showing that tailored hybrid architectures offer practical solutions for resource-constrained environments and low-resource languages. Acknowledging dataset size, hardware limitations, and restricted task scope, future work should expand evaluations to larger datasets, additional languages, and more advanced model variants.

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